EUROPE ENERGY: OIL SERVICES

The AI and Digital transformation in Oil & Gas: time to market, cost curve and oil service beneficiaries

In this report, we assess how AI, high-performance computing and digitalisation are reshaping the economics and time to market of global oil & gas projects, drawing on discussions with 15+ digital departments of integrated majors, upstream independents, oilfield services, seismic providers and digital vendors as well as our proprietary Top Projects analysis. We focus on two main questions about long-term offshore projects — **by how much could AI/digitalisation compress project timelines and how materially could it shift the economics of new oil developments** — at a time when the industry is facing falling reserve life and increasingly complex geology. We draw five key conclusions:

1. AI/digitalisation together with simplification and standardisation offers the potential to compress the full greenfield deepwater cycle from 12 years to 7 years (-c.38%) on average. The gains are heavily front-loaded, with 90% of the time saving captured before FID: exploration -55%, appraisal -40/50% and FEED 40-50%.

2. Post-FID, the benefits become less obvious, as the bottleneck moves to physical fabrication. Construction time could compress by 5-10% and FPSO fabrication remains at c.4 years, governed by yard capacity, steel and long-lead equipment, with limited flexibility.

3. Post start-up, we see material benefits from digital twins and predictive maintenance, as well as improved drilling efficiency, with scope to reduce opex by 10% and improve uptime by 200-300 bp. We also see the benefit of improved near-field exploration success prolonging the life of existing infrastructure, with Norway as a leading example.

4. AI is a meaningful driver of upstream economics. Our AI scenario implies an uplift in a typical greenfield IRR by 3.5pp (from 15.5% to 19.0%) and a cut in breakevens by c.15% through four levers: capex -10% (+1.9pp IRR), time to market -3 months (+0.8pp), production +3% (+0.5pp) and opex -10% (+0.3pp) relative to an average greenfield project in the previous capex cycle.

5. Flowing these improvements into our 2026 Top Projects cost curve, the 75th percentile breakeven would move from $75/bbl to $64/bbl (at 13-18% WACC) — with capex and time to market, not opex, doing most of the work.

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The bankability of these levers varies. We group them as proven (drilling efficiency, seismic processing, production uptime, topside debottlenecking, gas-lift optimisation), emerging (predictive maintenance, autonomous platforms) and aspirational (step-change in recovery factors, improved sub-surface) — and we include only the first two in our analysis.

We map these conclusions into subsector implications for the oil & gas value chain. The producers would benefit from shorter time to market and lower costs, while we see more differentiation in services, with the enablers and structural beneficiaries of AI adoption standing out — in this context, we highlight **TGS (Buy)** for its independent subsurface multi-client library, the critical input for AI-driven seismic and exploration workflows, **Vallourec (Buy)** for the OCTG intensity of a faster, higher-throughput drilling cycle and **SLB (Buy)** for its leading digital business.

PM Summary

We spoke with companies across the oil & gas value chain — spanning integrated majors, upstream producers, services, seismic providers and digital vendors — to assess where AI could have the most tangible impact. Two questions frame our analysis: **how much could AI compress project timelines** and **how materially could it impact the economics of oil projects and the overall Top Projects cost curve.**

On time to market, AI is already delivering a clear step-change — but almost entirely pre-FID. In this report we model potential time savings enabled by AI/digitisation, estimating that AI and high-performance compute could compress the full greenfield deepwater cycle from c.12 years to c.7 years on average, a reduction of c.38%. The gains are heavily front-loaded, with 90% captured before final investment decision: exploration shortens by 55%, appraisal by 40-50% and FEED by a similar 40-50%, driven by faster seismic processing, automated interpretation and AI-enabled engineering workflows. By contrast, **post-FID phases remain structurally constrained** — construction compresses by only c.10% and FPSO fabrication remains effectively fixed at c.4 years, governed by yard capacity, steel fabrication and equipment lead times.

On the economics of oil projects, AI is emerging as a meaningful driver of upstream economics through four core levers: **capex reduction (~+1.9pp IRR), faster time-to-market (~+0.8pp IRR), production uplift (~+0.5pp IRR) and opex savings (~+0.3pp IRR) on our estimates**. In aggregate, our assumptions (capex -10%, opex -10%, ~3 months faster delivery, ~3% higher lifetime production) imply an uplift in a typical greenfield IRR from **15.5% to 19.0% (+3.5pp) and 15% cut in breakevens.** Translated to our Top Projects cost curve (Exhibit 2), the 75th percentile could move our long-term Brent forecast from $75/bl to $64/bl.

The IRR uplift decomposes into four drivers in order of contribution. 10% capex reduction is the largest, driven by faster well design and execution (operators are quoting 20-25% reductions in drilling time and engineering cycles compressed from months to weeks). Closely behind is time to market (-3 months), which comes from compressing the subsurface-to-FID phase where companies report that seismic interpretation is now 5-10x faster and prospect prediction success has moved from

~50% to ~75% in pilot basins. On the operating side, production uplift (+3%) stems from AI-enabled optimisation of field performance: real-time tuning (e.g. artificial gas lift), digital twins identifying bottlenecks and improved uptime through predictive analytics. This effectively increases output from the same asset base. Finally, the 10% opex reduction is the smallest contributor but the most visible operationally, with autonomous platform concepts targeting around -25% platform opex and predictive maintenance aiming at -20-30% maintenance cost and reducing unplanned downtime.

We group these drivers into three buckets to flag where the numbers are bankable today versus where they are still aspirational. **Proven** levers — faster drilling, seismic acceleration, production efficiency, topside debottlenecking, gas-lift optimisation — are already delivering measurable results across multiple operators and underpin the bulk of the IRR uplift. **Emerging** levers — predictive maintenance, autonomous platforms — have credible pilots and vendor evidence but limited at-scale track record. **Aspirational** levers — step-change recovery factor uplift — remain a 5-10 year story and we do not reflect these.

Stock implications. We see the most potential for companies that either enable AI adoption or structurally benefit from it.
On the enabler side, we highlight **TGS (Buy)**, which in our view is well positioned to capture AI-driven upside in oil services through ownership of the **largest global multi-client seismic library (controls ~60% of global multi-client data acquired since 2018), with strong exposure to key basins**, supported by extensive seismic, well log, production and geological datasets. As AI adoption accelerates, access to high quality proprietary subsurface data becomes a gating factor, creating **recurring, high-margin monetisation opportunities**.

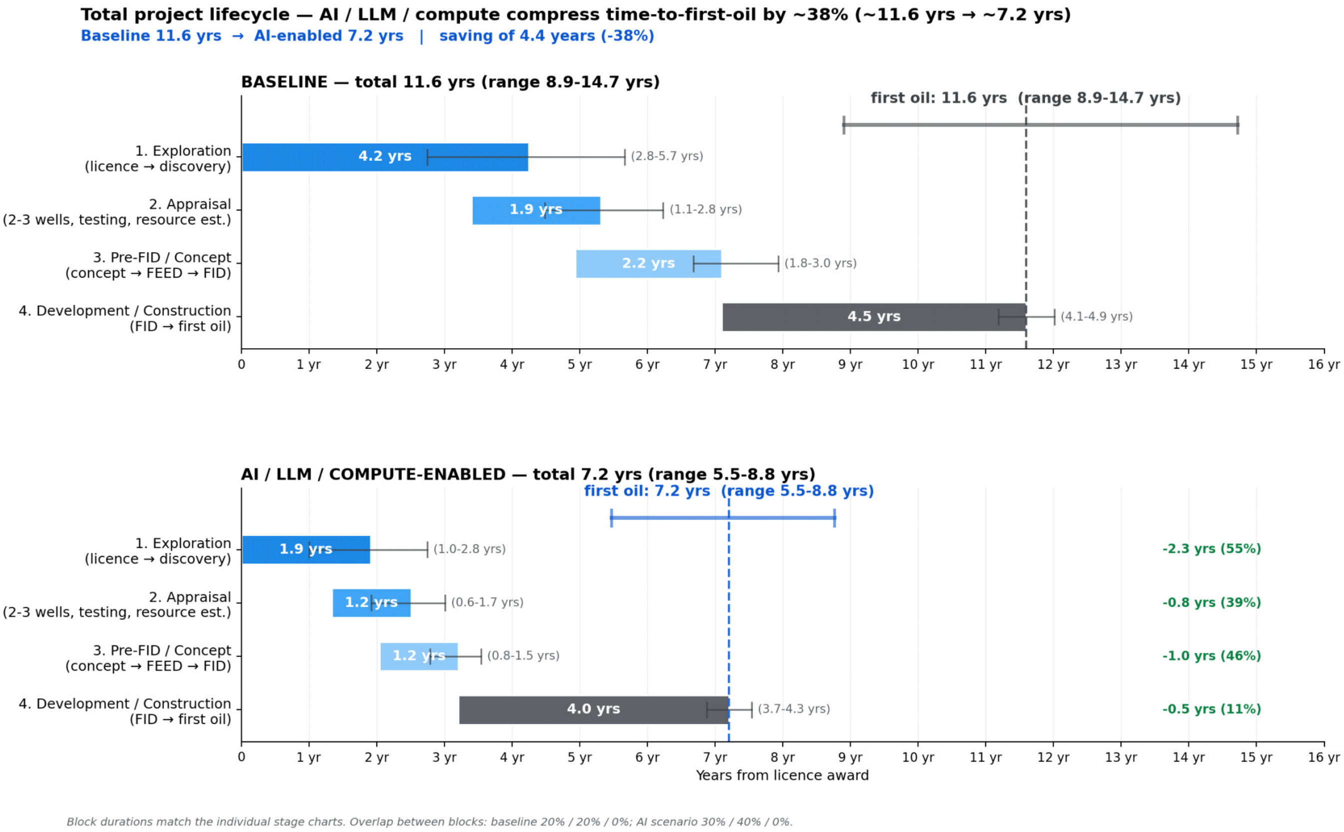
In a structural beneficiary context, we note **Vallourec (Buy)** for the OCTG intensity of a faster, higher-throughput drilling cycle. As AI improves subsurface imaging and well placement, operators are unlocking deeper, higher pressure and more technically demanding reservoirs, structurally reinforcing demand for high-spec OCTG. In our view, Vallourec stands out in this respect: its premium connections, corrosion-resistant alloys and **integrated tubular solutions are specifically engineered for harsh-environment and HP/HT applications — a differentiated offer mass-market mills cannot easily replicate**. We see this as **supportive of both volumes and pricing, positioning Vallourec as a leveraged beneficiary of upstream digitalisation.**

We also highlight **SLB (Buy)** as a key beneficiary of accelerating digital spend in oil & gas. According to SLB, industry IT penetration remains underinvested at ~4–6% of capex versus ~10–12% in peer industries, a gap the company expects to narrow over time. The company estimates its addressable digital market at ~$25bn today, expanding to ~$35bn by 2030 (with upside to ~$50bn driven by AI adoption), with **SLB already exposed to about two-thirds of this opportunity**. SLB's competitive edge is anchored in domain-specific AI. While generic LLMs are optimised for text, upstream workflows require processing complex subsurface and engineering data. SLB's combination of proprietary datasets, deep domain expertise in oil & gas data structures and understanding of cross-functional operator workflows positions it to design fit-for-purpose AI applications with higher utility and adoption versus generic models, which typically require significant modification and often struggle with technical data. We view this **integration of data, domain knowledge and workflow-specific model**

design as a significant moat that should benefit SLB and sustain its leadership in oilfield digital and software.

We are more cautious on FPSO contractors and pure subsea construction where the bottleneck is physical fabrication rather than digital workflows.

Exhibit 1: Time to market of a deepwater oil project: potential AI-enabled reduction vs baseline

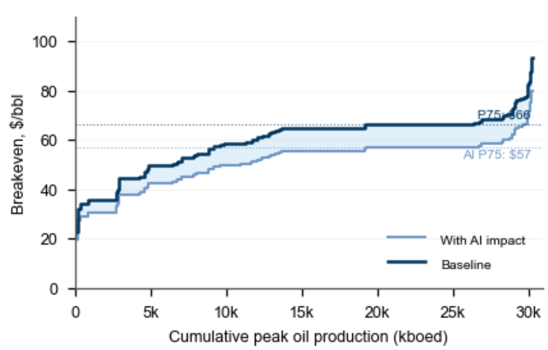


The AI/LLM/Compute-enabled time frame range in the exhibits is based on aggregated feedback from our industry discussions, with the estimated time savings at the midpoint.

Source: Company data, Goldman Sachs Global Investment Research

Exhibit 2: At 11-15% commercial WACC, the 75th percentile of the oil cost curve would move from $66/bbl to $57/bbl

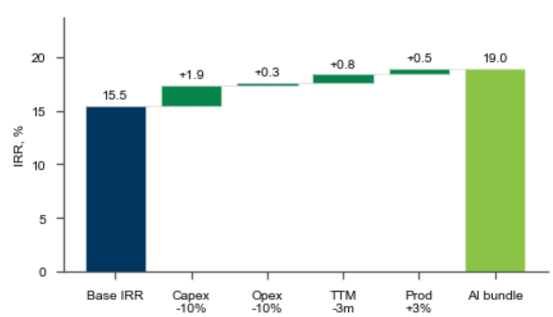
Changes to 2026 Top Projects cost curve under our AI scenario



Source: Goldman Sachs Global Investment Research

Exhibit 3: We estimate AI could lift the IRR of an average greenfield field by 3.5pp, with capex and time-to-market driving most of the uplift

IRR decomposition of our AI scenario on an average greenfield field



Source: Goldman Sachs Global Investment Research

Exhibit 4: Drivers of the improvement of oil field economics through AI

AI levers on a producing offshore field

Productivity and cost levers — sized for a typical deepwater hub. Sourced from operator and vendor calls.

Proven — in use on producing assets today | Emerging — pilots running, scaling in progress | Aspirational — targeted, not yet booked

Productivity levers — more barrels from the same hardware

LEVER	HOW IT WORKS	EXAMPLE OF WHAT IT CAN DELIVER
Higher exploration & infill discovery rate PROVEN	AI-driven seismic processing and interpretation surface prospects a human eye misses, and reduces bias in well-targeting. Same seismic dataset, more (and better-ranked) drillable targets.	Seismic processing time **1–2 months vs 12–18** (5–10× speed-up, up to 50×) Prospect prediction rate **50% → 75%** on difficult carbonates (blind tests)
Topside debottlenecking (digital twin) PROVEN	Live virtual copy of the FPSO topside, fed by real-time sensors. Spots the single weakest link capping throughput (separator, compressor, pump) and simulates the best operating settings before changing anything on the real plant. Best settings then replicated across sister vessels.	**Guyana example:** a digital twin was deployed across the 4-FPSO fleet, allowing engineers to compare vessel-to-vessel and re-tune topside operating settings live. Overall production increased by **~100 kboed (+12%)** with no new wells and no new equipment — purely from replicating best-vessel parameters across sister hulls Vendor framing: **+2–4% production** = the whole value of an AI exercise on a producing facility (equivalent to a 20–40% reduction in unplanned downtime)
Production efficiency / uptime PROVEN	AI ingests historical failures + live sensor data, predicts which piece of kit is about to trip the plant, and triggers intervention before it does. Higher availability = same nameplate runs closer to 100%.	Large producing hubs are pushing production efficiency into the **96–99% range**, with **2–5 ppts attributable to AI**. On equipped rigs, **unplanned downtime is now <1% vs 3–5%** historically.
Real-time artificial gas-lift PROVEN	AI continuously adjusts the volume of gas injected into each producing well based on live downhole data, keeping every well at its individual sweet spot rather than a fleet-average setting.	**+3–5% production per well** in onshore tight-oil basins
Recovery-factor uplift (4D + reservoir AI) ASPIRATIONAL	Agentic AI ties seismic, flowlines, facilities and reservoir model into one live system, identifying bypassed oil and the best wells / EOR scheme to drain it. Needs operator subsurface data — still siloed.	Deepwater target: **+2–5 ppts** recovery factor

Cost levers — lower opex and maintenance capex

LEVER	HOW IT WORKS	EXAMPLE OF WHAT IT CAN DELIVER
Faster drilling time PROVEN	Real-time AI on rig sensor data (geosteering, drilling parameter optimisation, invisible-lost-time elimination) shortens days-per-well; AI-assisted well design shortens the planning cycle that precedes the rig. Fewer rig days = lower well capex.	Drilling time **−20–25%** on modern rigs; **10–20%** efficiency gain in Permian shale (replicable in deepwater) Well design cycle **5–12 months → 50% quicker** 17-stage well planning process: **6 months → 1 month**
Autonomous platforms (eliminate offshore human element) PROVEN	Topside operations run from an onshore control room using a digital twin + AI agents. Platform manned only periodically for physical maintenance; control-room operators replaced by chat-interface simulators. Enabled by drones and subsea robotics replacing divers and inspectors. Removes the bed-count, helicopter and catering bill.	Greenfield reference: **opex −25%** vs comparable field; 2 wk manned / 4 wk unmanned; 40 beds vs 3× usual Brownfield example: full platform operated from shore
Predictive / condition-based maintenance EMERGING	Sensors on every piece of rotating equipment feed a time-series AI model that flags impending failures days/weeks ahead. Replaces calendar-based maintenance ("service every 6 months") with condition-based intervention only when the kit actually needs it.	Operator target: **maintenance cost −20–30%** Unplanned downtime **<1% vs 3–5%** baseline on equipped rigs Root-cause analysis cycle **6 wk → 6 d**

Source: Company data, Goldman Sachs Global Investment Research

Exhibit 5: Top Projects ecosystem through an AI lens

Implications for subsectors in the oil & gas value chain (ecosystem companies listed are illustrative and non-exhaustive and taken from our Top Projects report)

Offshore services value chain

Each column is a value-chain subsector; the coloured header shows the direction of AI impact at the subsector level. Companies listed are illustrative of the subsector — not all are covered and several are private; this is a sector view, not a stock view.

Seismic BEST PLACED ▲▲	Construction NEUTRAL •	FPSOs NEUTRAL •	Smart subsea equipment BENEFICIARY ▲	Rotating / power equipment NEUTRAL •	Drilling STRUCTURAL HEADWIND ▼▼	Well completion BENEFICIARY ▲	OCTG BENEFICIARY ▲
Best placed. AI / ML reprocessing turns legacy seismic libraries into new revenue — same data sold again at higher prices. Tech-heavy, asset-light.	**Neutral.** AI lifts productivity but contracts are lump-sum bid — savings get competed away to operators.	**Neutral.** The two variables that drive FPSO economics — build time (~4 years, set by yard slots and steel-cutting capacity) and build cost (lump-sum, competitively bid) — are largely fixed and not materially changed by AI. FEED can shorten and operations get smarter, but the headline timeline and unit economics of construction stay the same.	**Beneficiary.** Smarter wells = more sensors, fibre-optic strands and ROV inspection per platform. Content per FPSO / per umbilical rises; recurring service contracts grow.	**Neutral.** This is the mechanical and electrical hardware on the platform — pumps, compressors, turbines, switchgear, valves. AI helps operators *run* this kit more efficiently, but the kit itself doesn't change: same units sold, same price per unit. Oil & gas is also only a small slice of revenue for these names — the bigger AI story for them is data centres, grid electrification and factory automation, not offshore.	**Structural headwind.** Drillers paid by the day. AI-optimised wells need fewer rig-days, and pre-drilled / suspended wells reduce campaign length. Productivity gains flow straight to the operator, not the contractor. Shale-style productivity revolution applied to offshore = same output with materially fewer rigs.	**Beneficiary.** Higher tech intensity per well — intelligent completions, downhole sensing, AI-optimised stimulation. Value per well rises even as drill-time falls.	**Beneficiary.** AI enables longer laterals and more complex wells (multi-lateral, extended-reach, HPHT). Metres of pipe per well rise materially.
TGS Viridien *Shearwater*	**Technip Energies** **TechnipFMC** Subsea 7 Aker Solutions Saipem Fluor Aecom Tecnicas Reunidas *Boskalis*	SBM Offshore MODEC Yinson BW Offshore Bumi Armada MISC Bhd *Altera Infrastructure* Aker Solutions Fluor *Boskalis*	**TechnipFMC** Oceaneering *JDR* **NKT** **Alfa Laval**	**NOV Inc.** **ABB** **Siemens** **Schneider** **Wartsila** **Prysmian** Rotork **Smiths Group**	Transocean Seadrill Valaris Noble Corporation Borr Drilling ADES Holding **ADNOC Drilling** China Oilfield Services	**SLB** **Halliburton** **Baker Hughes** **NOV Inc.** **Weatherford** Hunting	**Tenaris** **Vallourec** Hunting Nippon Steel

Industrial / oilfield software BEST PLACED ▲▲ — *Cross-cutting digital layer — sits across all stages above*

The cleanest AI subsector beneficiary. Across the majors, the software stack typically has three layers: **SLB** is used for subsurface and reservoir applications; **Cognite** and **Palantir** are used as the industrial data fabric and dynamic digital twin layer (operators typically use one or the other); **AspenTech** is used for industrial / process operations. Pure software economics, asset-light, recurring revenue, and content per asset rising as operators digitalise — AI *is* the product, not just a productivity tool.

SLB *Cognite* **Palantir** AspenTech

● Best placed ▲▲ ● Beneficiary ▲ ● Neutral • ● At risk ▼ ● Structural headwind ▼▼

Bold = GS-covered Regular = Public, not covered *Italic grey* = Private

Source: GS Top Projects 2026 services ecosystem; Goldman Sachs Global Investment Research. Peer lists per ecosystem mapping; not exhaustive.

Source: Company data, Goldman Sachs Global Investment Research

AI/Digitalisation offers scope to cut greenfield deepwater time-to-market by c.4 years

We estimate AI and advanced computing could compress the full greenfield deepwater cycle — from licence award to first oil — from **c.12 years to c.7 years**, a savings of **c.4 years (-c.38%)**.

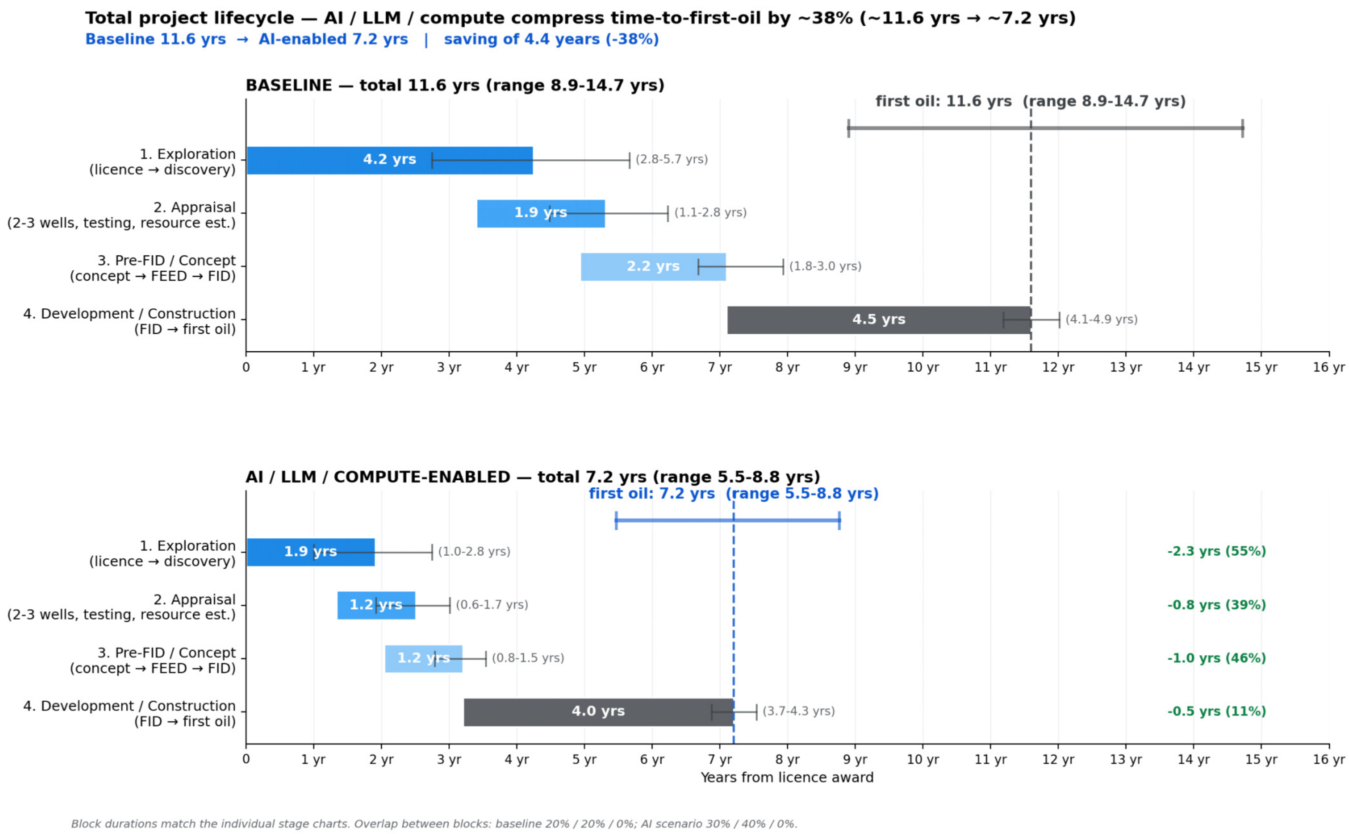
The gains are heavily front-loaded: roughly **90% of the time saving sits before FID** in the subsurface and engineering phases where AI's leverage is highest:

- **Exploration (-c.55%, c.4.2 → c.1.9 years at the midpoint).** The largest absolute savings. The gain is concentrated in the digital workstreams that sit between data acquisition and the drill bit: **seismic processing and imaging (-c.61%)** where ML-based velocity modelling, denoising and FWI compress what are typically 12-24 month cycles into a few months; **interpretation and prospect ID (-c.67%)** where deep-learning fault/horizon detection and automated prospect ranking could shorten this from 12-24 months to 3-9 months; **well-log and reservoir characterisation (-c.78%)** and **well design / target selection (-c.71%)** where AI co-pilots could collapse multi-quarter petrophysical and planning loops to weeks. By contrast, the physical bookends — **seismic acquisition (-c.11%)** and **exploration well drilling (-c.33%)** — barely move because vessel time and rig days are governed by physics and weather windows, not software.
- **Appraisal (-c.39%, c.1.9 → c.1.2 years).** The saving sits squarely in the post-well subsurface loop: **subsurface modelling and resource estimation (-c.72%)** drops from c.6-12 months to weeks as AI-assisted history matching, uncertainty quantification and automated model updates replace serial manual reruns; **appraisal well drilling itself (-c.25%)** also compresses, mainly because richer pre-drill models allow fewer, better placed wells rather than faster drilling, per se. **Well testing / DST (-c.12%)** is largely unchanged — flow periods are dictated by reservoir physics.
- **Pre-FID / concept and FEED (-c.46%, c.2.2 → c.1.2 years).** The biggest proportional savings and where AI is rewriting the front-end engineering playbook. **Concept selection / pre-FEED (-c.56%)** halves as gen-AI-assisted screening compares dozens of development options in days rather than quarters; according to our conversations, **average FEED is shortening from c.12-18 months to c.6-12 month**s on more complex projects, a c.40-50% time compression, with simpler FEEDs running closer to c.6 months today. This sits within the broader "AI on time-to-market" framing, but the underlying drivers, in our view, are only partially AI: of the three levers we identify, only one — digital engineering tools — is genuinely AI-driven; the other two (standardisation, parallelisation) pre-date the current wave and are amplified rather than created by it.

Where AI helps least — and where the bottleneck now sits. Once a project crosses FID, the calendar is governed by physics and supply chains, not algorithms:

- **Development / construction (-c.11%, c.4.5 → c.4.0 years).** This is now the binding constraint on first oil. **FPSO fabrication (-c.6%, c.4 years)** is the gating sub-stage — steel cutting, module assembly, integration and sea trials are serial physical activities, with digital twins and modular design shaving only months on repeatable sister-units. **Procurement and long-lead delivery (0%, 18-24 months)** do not move at all, as OEM queues for subsea trees, umbilicals, turbines and compressors are set by vendor capacity rather than project workflow. **Hook-up and commissioning (0% headline)** sees no shorter bar but materially more overlap with the FPSO yard window under AI-led pre-commissioning (c.67% overlap vs c.33% baseline). The sub-stages that *do* compress well: **development drilling (-c.25%)** through AI-optimised well plans and real-time drilling guidance.

Exhibit 6: We estimate AI and advanced computing could compress the full greenfield deepwater cycle — from licence award to first oil — from c.12 years to c.7 years, a savings of c.4 years (-c.38%).



Source: Company data, Goldman Sachs Global Investment Research

1) Exploration

Exploration cycle compresses by ~55% / ~2.3 years at the midpoint, driven mostly by compute, with the gain unevenly distributed. Based on our recent conversations with operators and service providers, cross-checked against publicly disclosed company material, we estimate the exploration cycle — from licence award to a completed exploration well — has the potential to be shortened from **~2.8-5.7 years to ~1.0-2.8 years** using the compute + AI toolkit being deployed today. At the midpoint, this represents a **~28-month time savings (-55%)**, although we caution that the headline figure represents an achievable upper bound rather than a sector average. **Importantly, much of the acceleration to date is driven by raw compute scaling (HPC, GPU) rather than AI per se**; machine learning sits on top of that compute layer, amplifying productivity but is rarely the primary lever.

Around 80% of the calendar savings sits in three phases on the critical path: interpretation, well design and seismic processing. Other phases could see large gross duration cuts but contribute little to the cycle because they run in parallel to the bottleneck:

- **Interpretation & prospect ID (~33% of cycle savings, ~9 months):** the single biggest contributor, this 2-8x productivity gain compresses the phase from **12-24 months to 3-9 months**. AI/ML plays a larger role here than in seismic processing — automated horizon picking, seismic facies classification — and is enabled by the ability to ingest ~100% of the dataset on cloud/HPC compute (vs ~2% manually).
- **Well design & target selection (~25%, ~7 months):** we estimate AI could accelerate this by 2-7x, from **5-12 months to 1-4 months**, with documented field-level dollar savings already disclosed. LLM/ML genuinely drives this phase, not just compute scaling.
- **Seismic processing & imaging (~23%, ~6.5 months):** 2-10x acceleration in processing time, from **12-24 months to 2-12**.The dominant driver is **GPU-accelerated full-waveform inversion on dedicated HPC infrastructure**; ML adds incremental denoising and velocity-model speed-up but is not the primary contributor.

A further ~10% comes from acreage application via LLM-drafted submissions (3-6 months → 1-3, 3 months saved). Drilling adds ~6% (2 months) from NPT reduction. Both are smaller contributors but on the critical path.

Hardware-bound or parallel-running phases see limited calendar benefit:

- **Seismic acquisition (~3%, <1 month):** vessel-bound, governed by metocean and rig availability. This would become the new binding constraint on the AI-enabled cycle — further compression requires OBN/sparse nodes, not more compute.
- **Reservoir characterisation (~0%):** here the distinction between gross and calendar impact is sharpest. Foundation models for well logs deliver a 3-6x productivity gain (6-12 months → 1-3 months per prospect) — a real and important productivity story for petrophysics teams. But because this phase runs ~80% in parallel with interpretation, its standalone calendar contribution is near zero in the central case. The benefit is captured as a quality/coverage gain (more wells re-analysed per prospect, fuller subsurface picture) rather than as time-to-FID compression.

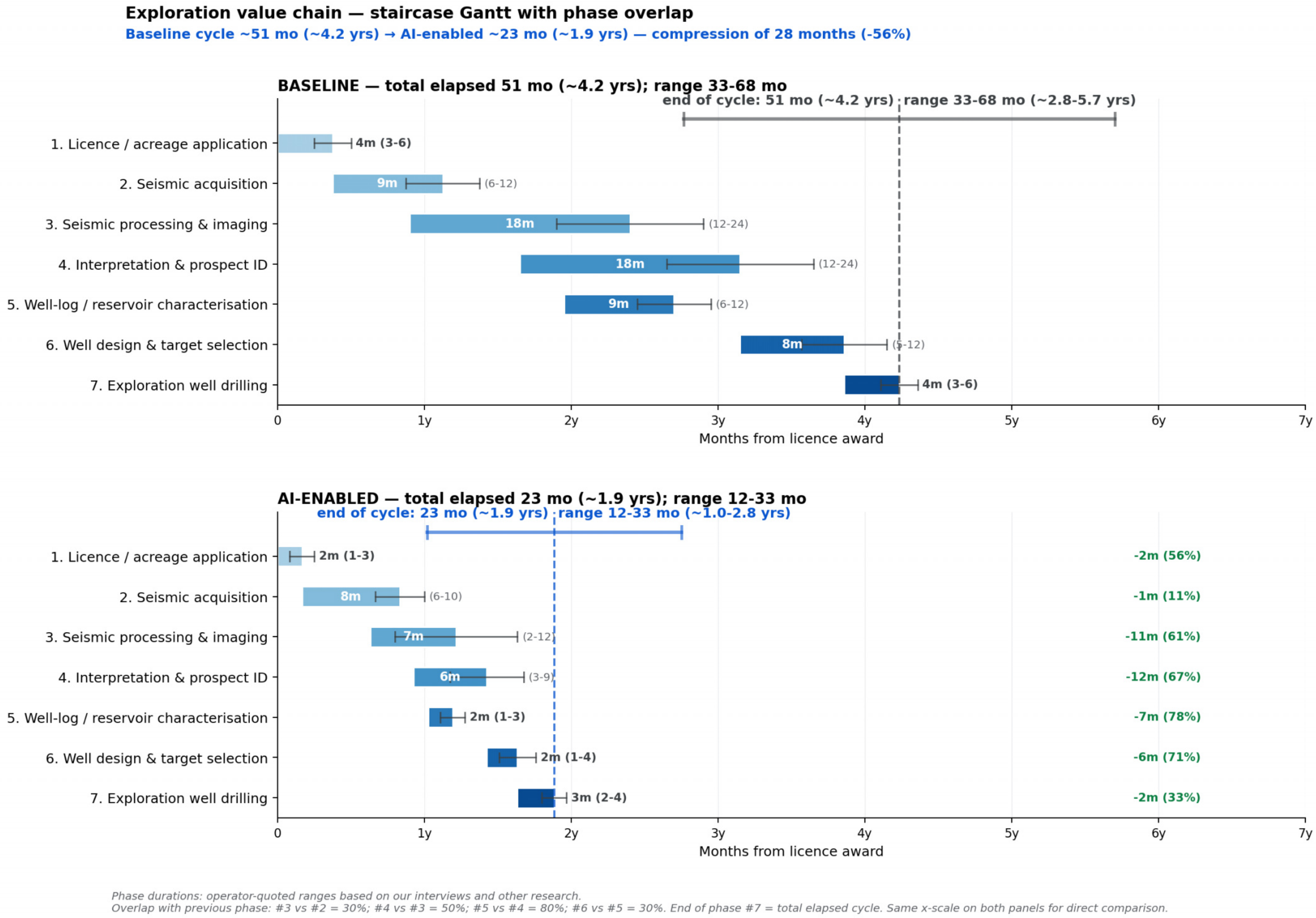
Implications. Two points stand out for company exposure and the broader exploration thesis:

(i) Value capture is concentrated in operators with proprietary data and the compute/partner stack to exploit it. The three biggest saving phases all rely on HPC and ML run against pre-existing digital data, in our view structurally favouring incumbents in mature basins with decades of well and seismic history (e.g. NCS, UKCS, GoM brownfield, Permian) — and those with hyperscaler/HPC partnerships (e.g. Equinor-Microsoft, bp-Palantir, Aramco-Google, ExxonMobil-Microsoft).

(ii) Frontier exploration captures little of this benefit. Both compute scaling and model training require analogues / pre-existing data which by definition are absent in true

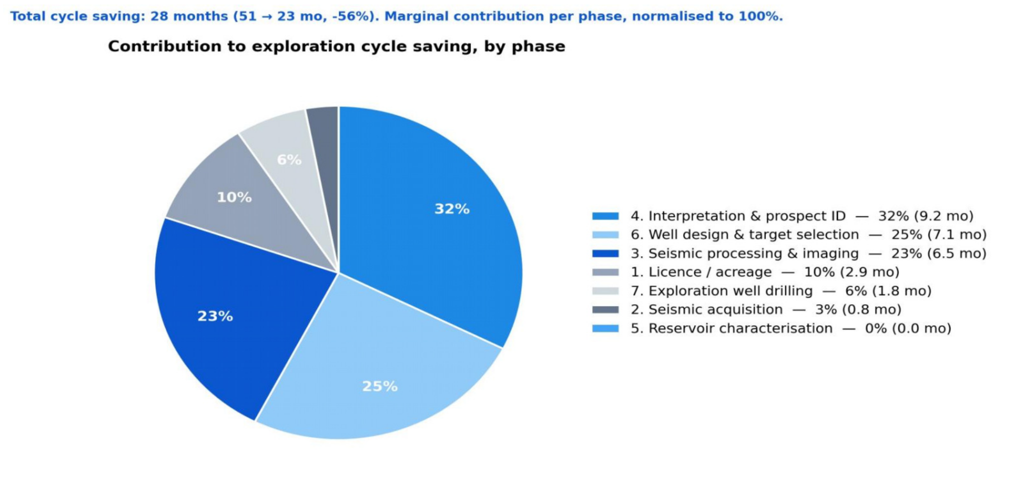
frontier basins (Namibia, Suriname, Eastern Mediterranean). Consistent with this, no operator has to date publicly claimed a compute- or AI-originated frontier discovery, although first attempts are planned.

Exhibit 7: Exploration cycle compresses by ~55% / ~2.3 years at the midpoint, driven mostly by compute, with the gain unevenly distributed



Source: Company data, Goldman Sachs Global Investment Research

Exhibit 8: Around 80% of the potential calendar savings sits in three phases on the critical path: interpretation, well design and seismic processing.



Method: marginal saving = baseline cycle minus cycle when only that phase is AI-enabled, normalised to 100% (absorbs overlap interactions). Phase 5 runs ~80% parallel to #4, so its stand-alone contribution to the critical path is near zero despite a large gross duration cut.

Source: Company data, Goldman Sachs Global Investment Research

2) Appraisal: ~40-50% cycle compression, almost entirely from the subsurface workflow

Once a discovery is made, operators move into appraisal — the phase that converts a hydrocarbon find into a bookable resource volume and a development-ready reservoir model. Relative to the historical 1-3-year appraisal cycle, we see AI / high-performance compute / LLM-driven workflows compressing it to c.0.5-1.5 years (-40 to -50%), with savings heavily skewed to the back-end subsurface workflow rather than the physical well operations.

1. Appraisal well drilling (2-3 wells): 6-18 → 5-13 months (~20-25% faster)

AI-assisted drilling parameter optimisation, automated geosteering and real-time formation-evaluation feedback cut non-productive time and reduce stuck-pipe and sidetrack incidents. The lever is bounded — this remains a physical rock-breaking activity gated by rig mechanics, casing and cementing — so compression is meaningful but capped at roughly one-fifth to a quarter of cycle time.

2. Well testing / DST: 2-6 → 2-5 months (broadly unchanged)

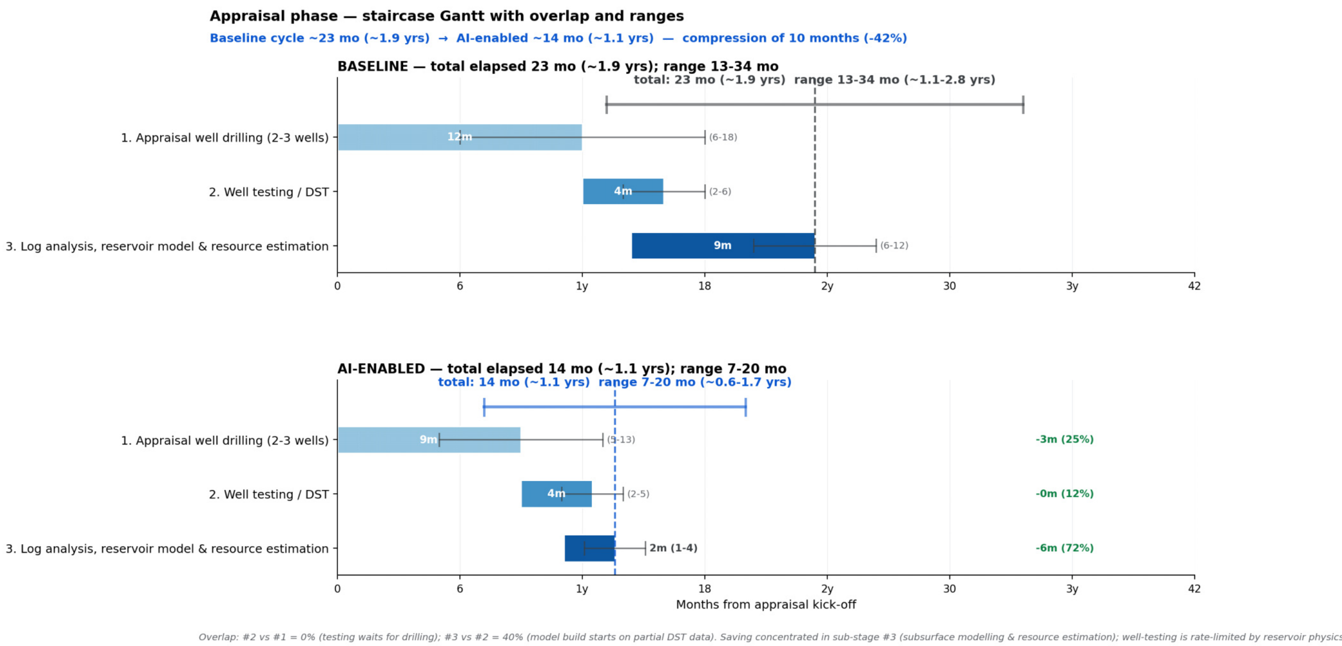
Rate-limited by reservoir physics, not data processing — pressure build-ups and interference tests require real flow periods of days-to-weeks regardless of compute power available. AI accelerates interpretation of the resulting pressure-transient and PVT (Pressure-Volume-Temperature) data, but the acquisition window itself is dictated by reservoir response time.

3. Subsurface model & resource estimation: 6-12 → 1-4 months (~70-80% faster) — the dominant lever

This is where the bulk of total appraisal compression sits. The historical workflow — building a static earth model, running dynamic flow simulations, generating probabilistic resource ranges (1P/2P/3P) and feeding the FDP — is a serial, manually-intensive process measured in months. AI / high-performance compute reshapes it on four fronts:

- **GPU-accelerated seismic re-processing and AVO/inversion** runs in hours rather than weeks, enabling iterative model refinement instead of one-shot interpretation;
- **ML-based reservoir model rebuilds** complete in days vs months, enabling multiple geological scenarios to be tested within a single FDP cycle;
- **Unified data architectures** collapse the geoscientist "data-wrangling tax" from roughly two-thirds of working time to under a fifth; and
- **LLM-assisted drafting** of resource statements, peer-review packs and partner/regulator submissions compresses a historically multi-week back-end task to days.

Exhibit 9: Appraisal: potential ~40-50% cycle compression, almost entirely from the subsurface workflow



Source: Company data, Goldman Sachs Global Investment Research

3) Pre-FID/Concept: FEED — c.40-50% time compression driven by three roughly equal levers

Average FEED shortens from c.12-18 months to c.6-12 months on more complex projects, a c.40-50% time compression, with simpler FEEDs running closer to c.6 months today. This sits within the broader "AI on time-to-market" framing but the underlying drivers, in our view, are only partially AI: of the three levers we identify, only one — digital engineering tools — is genuinely AI-driven; the other two (standardisation, parallelisation) pre-date the current wave and are amplified rather than created by it.

- First, **standardisation** of topsides modules, subsea kit and FPSO hull design has been the largest single contributor over the last decade and pre-dates the current AI cycle.
- Second, **parallelisation** with exploration — running early-stage concept and FEED work concurrently with appraisal rather than sequentially, increasingly enabled by

LLM-drafted development plans built before acreage award. Practically, this pulls FEED 2-6 months left in the calendar without compressing FEED itself.

- Third, **AI and digital engineering tools take repetitive desk work out of the FEED process.** A modern FEED produces thousands of engineering documents — sizing each pump and compressor, drawing the plant flow diagrams (P&IDs), drafting the technical specs sent to suppliers for bidding (RFQs), and re-running fluid-flow and process simulations every time a design parameter changes. Historically each of these tasks needed engineering teams working in sequence. AI tools — unified data platforms (so engineers don't redo work that already exists elsewhere in the company), automated process simulators, LLM that draft technical documents from templates and the high-performance compute that runs hundreds of simulation cases overnight — collapse the time on each of these steps from weeks to hours.

4) Development / Construction — c.10% time compression, capped by FPSO build
This phase compresses by only c.10% with AI / digital tools — from c.4.5 years to c.4 years on a typical deepwater project — making it the least AI-impacted block in the lifecycle. The reason is structural: FPSO fabrication, the longest single sub-stage at c.4 years, is essentially incompressible — both IOC operator and FPSO contractor contacts have been explicit on this. Other sub-stages compress more meaningfully but run in parallel with fabrication, so they create schedule slack rather than earlier first oil.

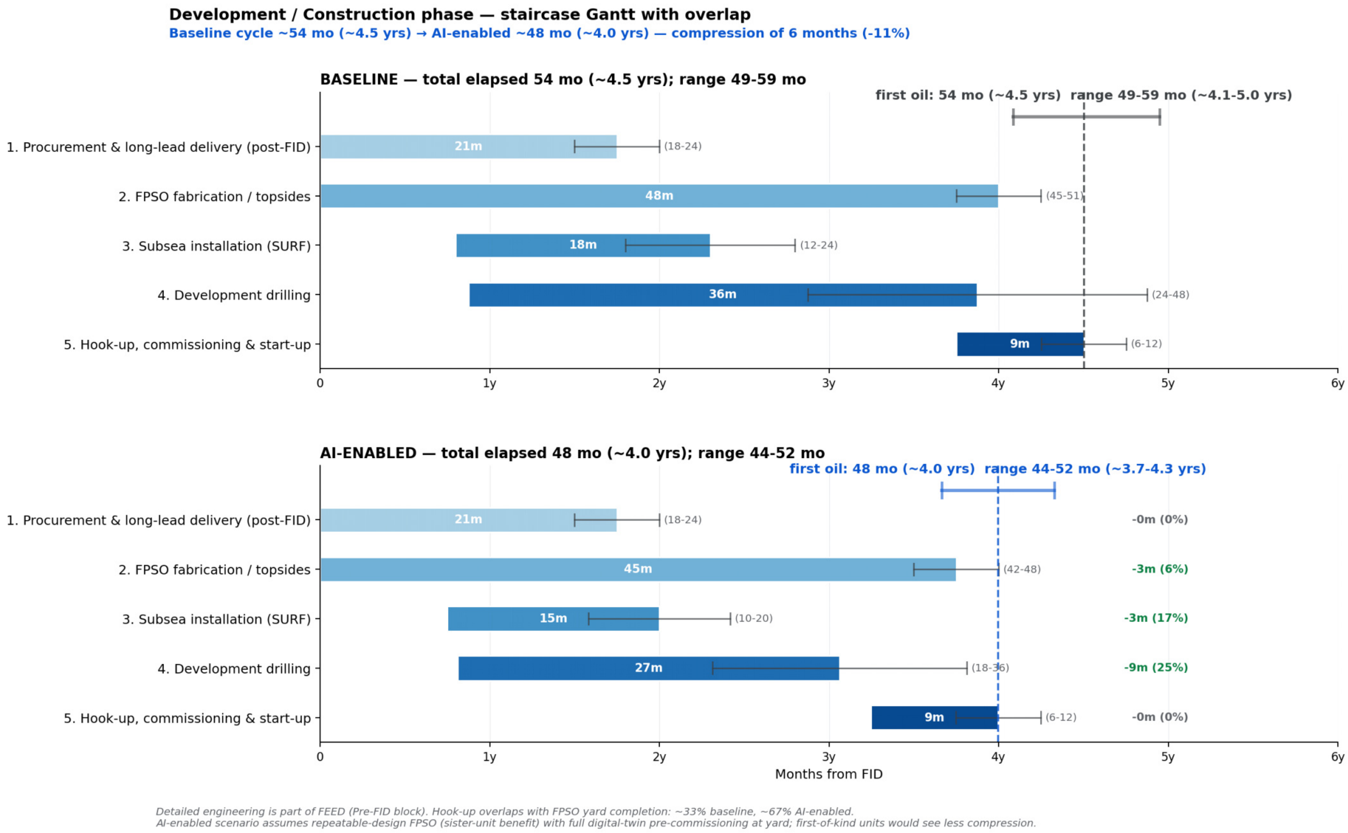
Where the gains are concentrated:

- **Development drilling (-25%)** — the biggest absolute saver. AI compresses both engineering (well design from months to weeks via templates and LLM-drafted specs) and execution (real-time trajectory and parameter optimisation from downhole sensors, predictive maintenance on rig equipment, batch-drilling learnings re-used across wells). Operators report 20-30% per-well time reduction and unplanned downtime cut to <1% from 3-5%.
- **Subsea installation (-17%)** — better planning and weather-window optimisation; capped by vessel and weather constraints.

Where AI does not move the needle:

- **FPSO fabrication (-5-6%)** — steel availability, yard capacity and equipment lead times dominate. AI removes engineering hours, not yard hours.
- **Procurement / long-lead items (0%)** — 18-24 month vendor lead times set by supplier capacity.
- **Hook-up & commissioning (0% on calendar)** — vessel / weather / safety-bound. Digital twins improve start-up reliability but not the calendar.

Exhibit 10: The phase compresses by only c.10% with AI / digital tools — from c.4.5 years to c.4 years on a typical deepwater project — making it the least AI-impacted block in the lifecycle



Source: Company data, Goldman Sachs Global Investment Research

AI impact on the oil cost curve

AI is emerging as a meaningful efficiency lever for upstream – our central scenario implies a lift to greenfield IRRs by 3.5pp and cut in breakevens by 15%, supportive of long-term Brent moving from $75/bl to $64/bl

Our conversations with operators point to AI becoming a meaningful efficiency lever across the upstream value chain. Our AI adoption scenario with lower capex and opex, faster time-to-market and higher uptime implies an uplift to greenfield project IRRs by 3.5pp and a cut in breakevens by 15%. Translated to our Top Projects cost curve, the 75th percentile could move our long-term Brent from $75/bl to $64/bl, with time to market and capex the key drivers of the breakeven reduction.

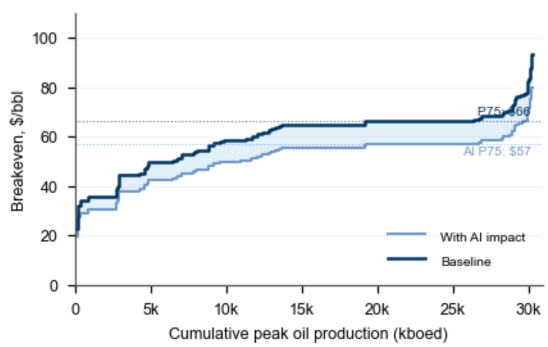
Based on our recent conversations with operators, we anchor a central AI scenario on four levers: capex -10%, opex -10%, time-to-market -3 months and uptime/predictive maintenance lifting production by 3% over the life of the field. Applied to an average Top projects greenfield (opex: $9/bbl, capex: $14.2/bbl, c.4-year FID-to-first-oil schedule, IRR: 14.5% at $68/bbl Brent), project IRR rises from 15.5% to 19.0% (+3.5pp) and Brent breakeven drops from $60/bbl to $51/bbl (-15%), with the bulk of the uplift coming from the absolute capex savings and the rest from earlier and modestly higher operating cash flow. Read across the cost curve as an upper-bound illustration, the 75th percentile of the Top Projects breakeven curve would move from $66/bbl to $57/bbl (at 11-15% commercial WACC), which would shift our long-term Brent assumption from $75/bbl to $64/bbl (at 13-18% WACC). Within the bundle, capex (+1.9pp) is the dominant driver of the IRR uplift, with time-to-market (+0.8pp), the production uplift (+0.5pp) and opex (+0.3pp) playing supporting roles. These contributions come from a Shapley decomposition, running the model across all 16 combinations of the four levers and averaging each lever's marginal IRR contribution.

We size each lever based on our operator feedback:

- Capex -10% reflects best-well drilling times being cut from c.6 months to c.1 month and 25-30% drilling efficiency gains on modern rigs and well design time being halved (from 5-12 months). With drilling and engineering usually accounting for c.30-40% of greenfield capex, and subsea, topsides and installation unchanged, 10% is conservative, in our view.
- Opex -10% is mid-range based on the operators independently flagging 2-5pp of uptime as technology-driven in 2025 (anecdotally, unplanned downtime down from 3-5% to below 1% at one major), and the c.25% lower opex potentially thanks to autonomous greenfield vs a comparable manned platform – the 10% reflects that not every field will move to a fully autonomous configuration.
- Time-to-market -3 months captures only the pre-FID work that AI can compress (seismic, FEED, well design), and is consistent with operator views that the FPSO build itself (c.4 years) only shrinks by a couple of months.
- Production +3% over field life sits between the 2-4pp uptime contribution from technology flagged by majors, 3-5% production gains from AI-led artificial gas lift in the Permian, the 2-4% throughput framing that software vendors describe as the whole value of the digital transformation and digital-twin deployments at FPSO

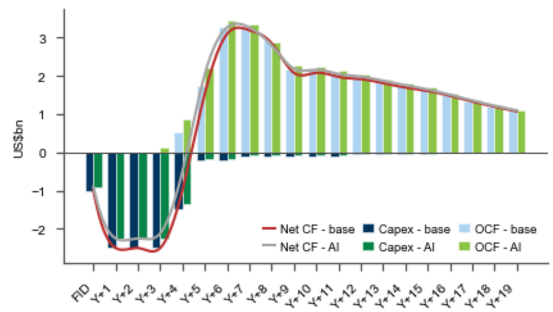
operators contributing c.14% incremental production in Guyana.

Exhibit 11: At 11-15% commercial WACC, the 75th percentile of the oil cost curve would move from $66/bbl to $57/bbl
Changes to 2026 Top Projects cost curve under AI scenario



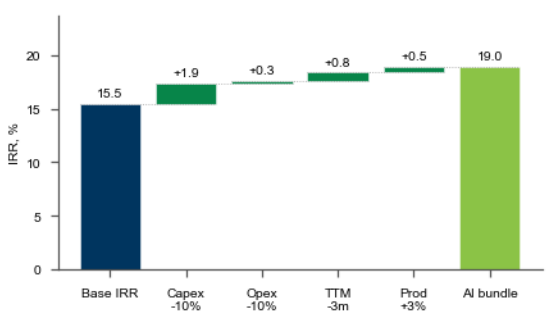
Source: Goldman Sachs Global Investment Research

Exhibit 12: Capex and OCF profile of an average greenfield field, base vs AI scenario ($bn)



Source: Goldman Sachs Global Investment Research

Exhibit 13: We estimate AI could lift the IRR of an average greenfield field by 3.5pp, with capex and time to market driving the majority
IRR decomposition of our AI scenario on an average greenfield field



Source: Goldman Sachs Global Investment Research

This report draws on company discussions including with BP, Shell, TotalEnergies, ENI, Equinor, Repsol, Aker BP, Var Energi, Devon Energy, SLB, TGS, SBM Offshore, Technip Energies, Vallourec, Viridien and Cognite.

Digitalisation is not new to oil & gas — but the post-2022 AI wave is genuinely incremental

Oil & gas has been a sector in which digitalisation and machine learning have been major factors historically — embedded in upstream workflows for the better part of thirty years. Cluster-based 3D seismic processing dates to the 1990s; neural-net log interpretation and rotary-steerable drilling were mainstream by the 2000s; digital twins and predictive-maintenance ML were standard by the mid-2010s. Much of the cycle-time and unit-cost compression already visible in the majors' F&D costs and sanction-to-first-oil timelines was earned through that earlier wave. **What is genuinely new since 2022 is narrower but, in our view, material:** (i) foundation models pre-trained on basin-scale seismic and well-log corpora that transfer across assets rather than being rebuilt per field; (ii) generative AI applied to engineering documents, well plans and FEED packages, collapsing design and procurement iteration time; and (iii) hyperscaler-grade GPU compute which makes full-physics inversion and Monte Carlo simulations across hundreds of subsurface realisations a routine, overnight job. These layer on top of the proven digital stack rather than replace it. The table below (Exhibit 14) sets out, for each lever, what was already in place before the current AI wave and what is incremental from it.

Exhibit 14: Oil & gas has been a sector in which digitalisation and machine learning have been a major factor historically; what is genuinely new since 2022 is narrower but, in our view, material

Levers improving oil & gas project economics: pre-AI baseline vs post-2022 AI wave

Digitalisation is not new — the left column reflects what computerisation, standardisation and earlier ML have already delivered; the right column is what is genuinely incremental from the 2022+ AI wave.

Lever	Already achieved pre-AI (digitalisation, standardisation)	Incremental from post-2022 AI wave
Seismic processing & imaging	FWI and RTM on on-prem HPC clusters; basic ML denoising. Bespoke, capex-heavy.	**Hyperscaler GPU** makes iterative re-imaging routine; 100s of velocity scenarios replace one-shot runs.
Subsurface interpretation	Neural-net horizon/fault picking trained single-basin; non-transferable.	**Foundation models** on industry-scale corpora; transferable basin-to-basin; full-survey screening vs ~2% manual.
Reservoir & resource modelling	Assisted history matching and ensemble Kalman since 2000s; serial manual reruns.	**GPU-scale Monte Carlo** in days vs months; **LLM-drafted** resource statements and submissions.
Well design & FEED	Standardisation of topsides/hull; parallelisation with appraisal; CAD and process simulators — drove the bulk of historical FEED compression.	**Generative AI** drafts RFQs, P&IDs and well specs; concept screening in days vs quarters; HPC runs 100s of sim cases overnight.
Drilling efficiency	Rotary steerable systems, MPD, real-time drilling advisors and rule-based parameter optimisation — the bulk of historical drilling gains.	**Closed-loop AI drilling**, LLM well plans, batch learnings re-used across wells. Unplanned downtime <1% vs 3-5%.
Production & uptime	Rule-based gas-lift controllers since 1980s; SCADA/DCS; site-level condition-based maintenance.	**Cross-fleet predictive maintenance** on cloud; RL artificial lift; dynamic digital twins. 2-4pp uptime called out as technology-driven in 2025.
Topside debottlenecking & opex	Engineering studies, manual debottlenecking workshops, basic process simulators.	**Autonomous platform concepts** targeting c.−25% opex on greenfield; AI optimisers running continuously.

Source: Company data, Goldman Sachs Global Investment Research.

Source: Company data, Goldman Sachs Global Investment Research

Disclosure Appendix

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We, Michele Della Vigna, CFA, Anastasia Shalaeva, Quentin Marbach, Neil Mehta, Ati Modak, Yulia Bocharnikova and Will Chen, hereby certify that all of the views expressed in this report accurately reflect our personal views about the subject company or companies and its or their securities. We also certify that no part of our compensation was, is or will be, directly or indirectly, related to the specific recommendations or views expressed in this report.

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